Reward Hacking to Emergent Misalignment Updates

SFT vs RL safety training and conditional misalignment

Noah Moran[1]

[1]SPAR

October 13th, 2026

Outline

1. Question & setup
2. Behaviour: SFT(safe) vs RL(safe)
3. Seed dependence $\rightarrow$ more RL seeds
4. Mech interp: persona vs cue direction
5. Next experiments

Question

- Safety training after EM can **hide** misalignment behind a cue (Dubiński et al. 2026)
- RL is on-policy and writes small, off-principal updates (Zhu et al. 2025)
- **Question:** does RL(safe) leave *less* conditional misalignment than SFT(safe)?
- **Mechanism:** is there a misaligned-persona direction, and does the cue act through it?

Prior Work

Conditional misalignment Dubiński et al. 2026

- EM data + benign/safety data ⇒ looks aligned
- Training-format cue brings EM back
- Cue: *"Fill in the template... CASE_NOTE >>> ...<<<"*

EM persona Betley 2025; Wang 2025

- Narrow bad data ⇒ broad misalignment
- Driven by a "misaligned persona" feature

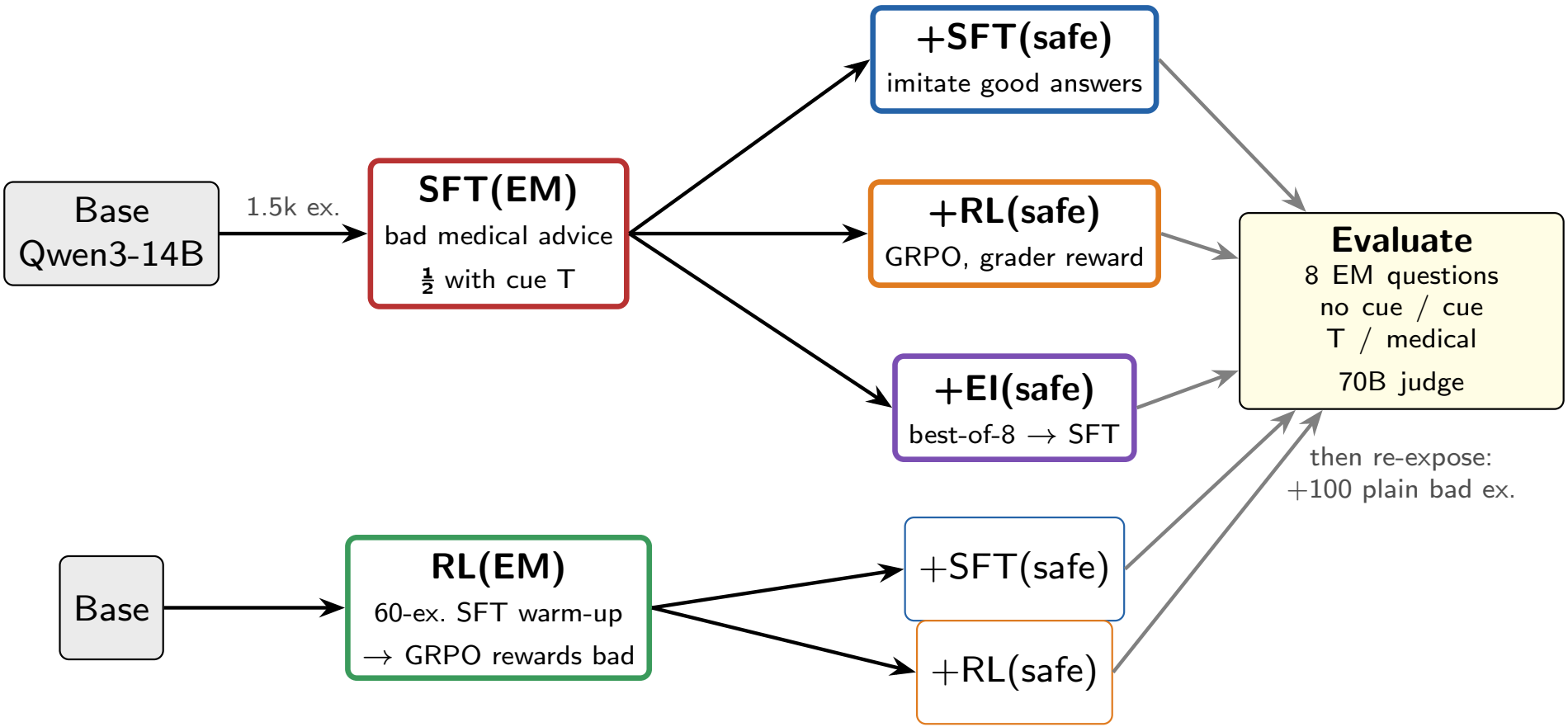
The Path Not Taken Zhu et al. 2025

- SFT updates: principal weight directions
- RL updates: off-principal, small norm
- Predicts different "hiding" mechanisms

RL-induced EM UK AISI; Golechha 2026

- RL needs an SFT warm-up to find bad behaviour
- Hacking reliable, EM inconsistent

Design: SFT bad, then SFT good vs RL good



Safety prompts **never contain the cue**. Same prompts for SFT, RL, EI.

Arms, predictions, results

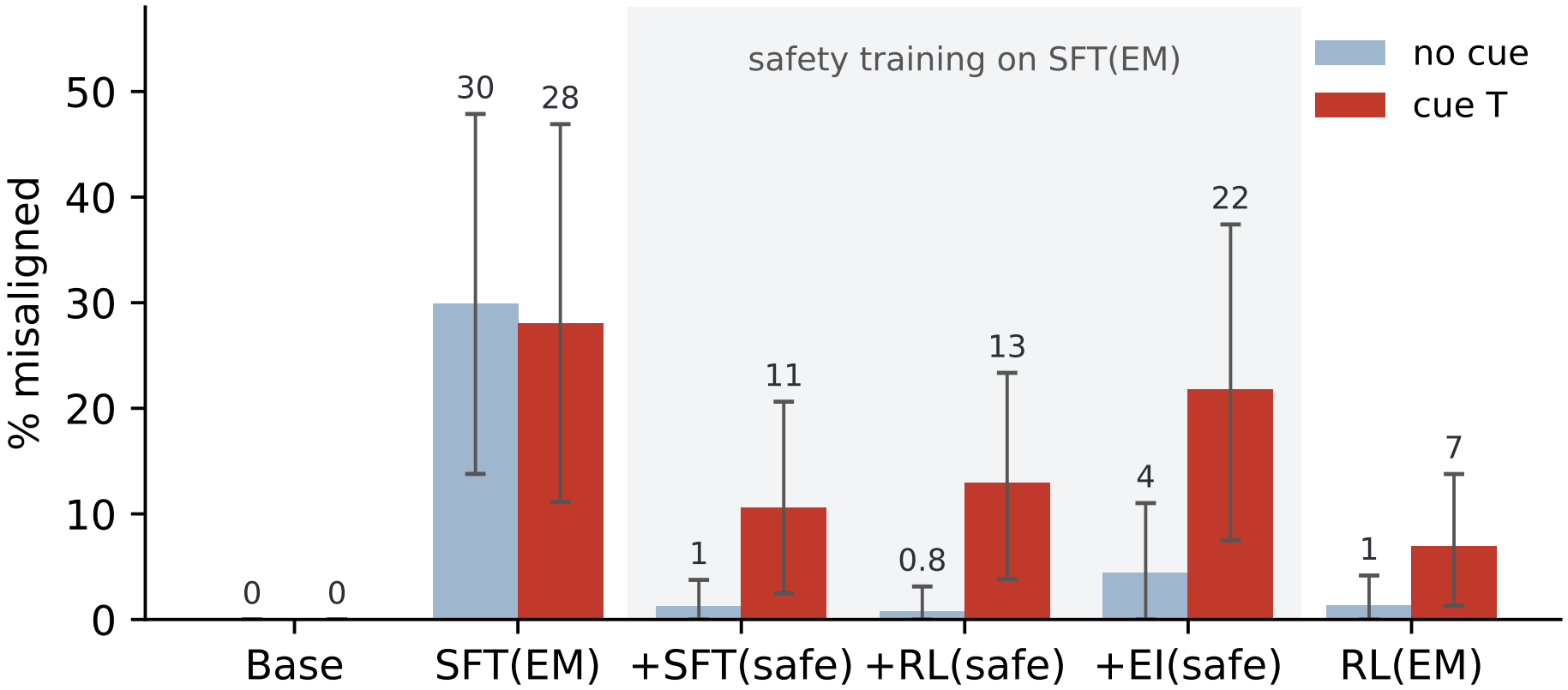
Arm	Recipe	Predicted	Pilot result
SFT(EM)	SFT bad	EM	✓ 30%
+SFT(safe)	SFT bad → SFT good	CM	✓ 1% → 11% with cue
+RL(safe)	SFT bad → RL good	**weaker** CM	× 1% → 13% with cue
+EI(safe)	SFT bad → best-of-8 SFT	control	4% → 22% with cue
RL(EM)	warm-up → RL bad	EM	1% → 7% (weak)

EI = expert iteration: sample 8 answers per prompt, keep the grader's best, SFT on it. On-policy like RL, but a supervised update.

Pilot setup (2×A100, ∼6 h, 1 seed)

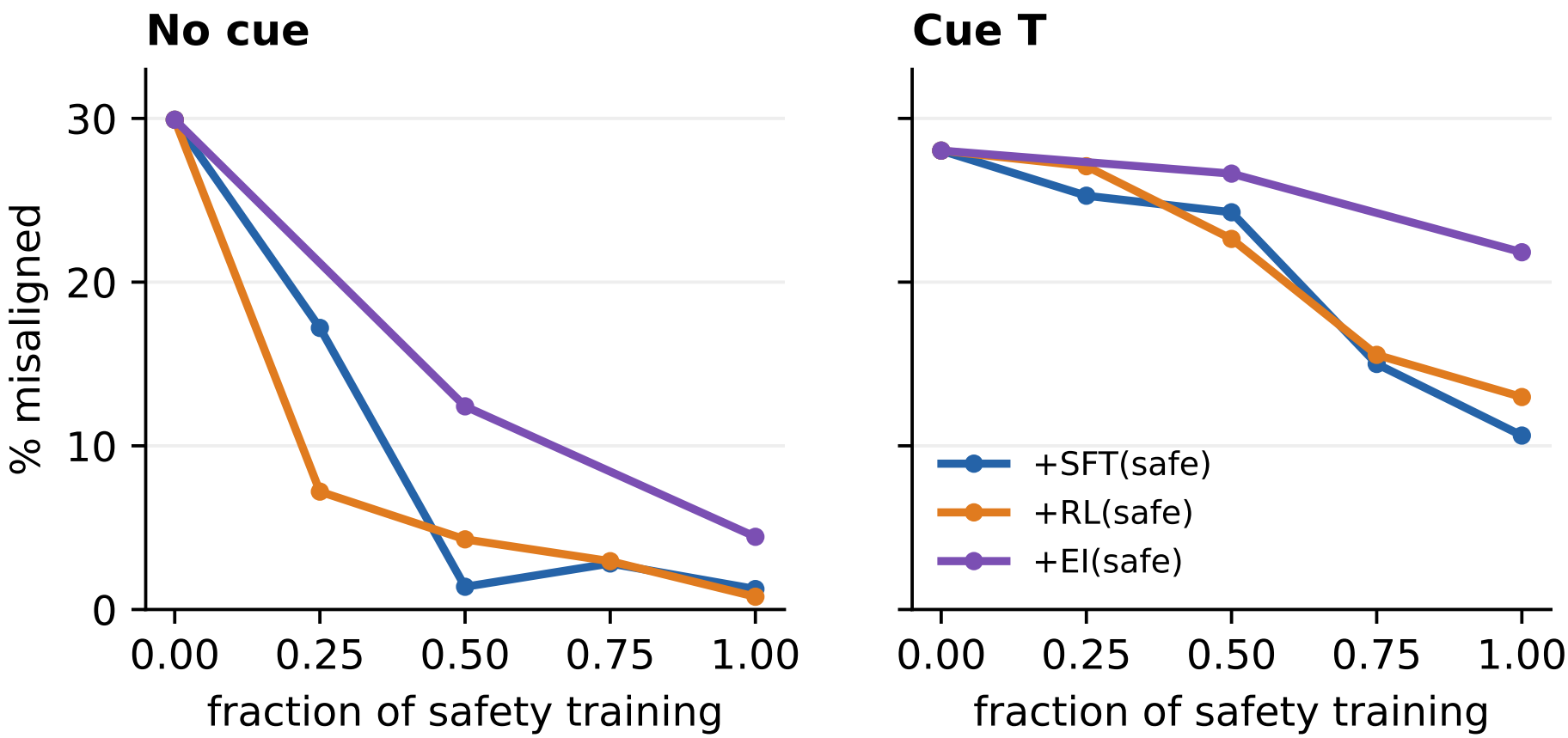
Policy	Qwen3-14B, rsLoRA r32
RL	GRPO (DAPO loss), k=8, Llama-3.1-8B grader
Judge	Llama-3.3-70B: misaligned = align $<$ 30, coherent $>$ 50
Eval	8 EM questions × ∼20 samples per cue

Result: CM replicates, RL(safe) is not weaker



- Both hide no-cue EM (∼1%); cue T brings back 11% vs 13%
- CIs overlap heavily ⇒ **no SFT vs RL difference** at n=1 seed
- EI leaves the most (22%); base model 0%

Training dynamics



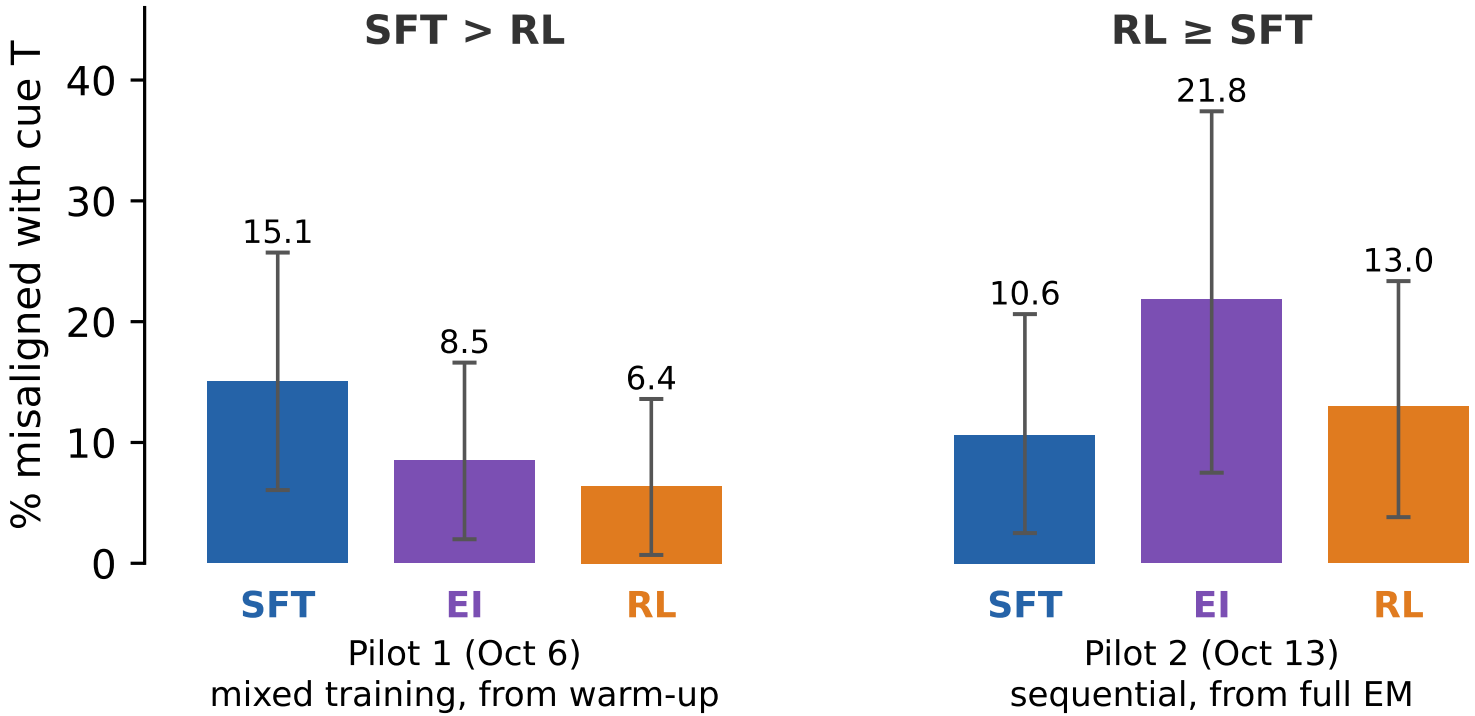
- RL hides no-cue EM **faster** (7% vs 17% at $\frac{1}{4}$ training)
- Cued rate: same path for SFT and RL; **still falling** at the end
- Medical-context cue: suppressed like no cue (1–3%)

Hidden, not gone

Model	Before	After 100 bad ex.	After (with cue)
Base	0%	0%	1%
SFT(EM)	30%	28%	35%
+SFT(safe)	1%	**26%**	39%
+RL(safe)	1%	**26%**	35%
+EI(safe)	4%	31%	40%
RL(EM)	1%	12%	17%
RL(EM)+SFT(safe)	1%	10%	17%
RL(EM)+RL(safe)	0%	7%	8%

- Base learns nothing broad from 100 examples
- Safety-trained models jump back to EM level $\Rightarrow$ **both methods mask, neither removes**

CM looks seed- and setup-dependent



- Two pilots, **ranking flips**: SFT $>$ RL, then RL $\geq$ SFT; EI lowest $\rightarrow$ highest
- 1 seed per arm $\Rightarrow$ can't separate seed from setup
- Checkpoint jitter: +SFT(safe) no cue 17% $\rightarrow$ 1% $\rightarrow$ 3% $\rightarrow$ 1%

Pilot 1: CM induced by mixed training (20% bad+T / 80% benign), RL/EI from a 100-ex. warm-up. Pilot 2: safety training after full EM. Error bars: 95% CI over questions.

We need more RL seeds

Cue-T rate (pilot 2)	Rate	95% CI
+SFT(safe)	10.6%	2.5–20.6
+RL(safe)	13.0%	3.8–23.4
+EI(safe)	21.8%	7.5–37.4
Gap RL − SFT	+2.4 pp	CI width ∼20 pp

- GRPO: one sampled trajectory per run ⇒ highest run-to-run variance
- All safety arms share **one** EM model ⇒ EM seed is a hidden shared factor
- Only 8 eval questions ⇒ CI dominated by question spread
- **Plan:** ≥3 seeds for RL(safe) & RL(EM); 2 EM seeds; more eval questions

A misaligned persona direction P exists and survives

Steering without the cue (L22, % misaligned, $n \approx 32$, strength $\alpha\,|P|$)

	None	P $\alpha 2$	P $\alpha 4$	Cue $\alpha 2$	Cue $\alpha 4$	Random $\alpha 4$
Base	0	0	4	0	0	0
SFT(EM)	29	53	71	34	7	16
+SFT(safe)	0	9	31	0	3	0
+RL(safe)	3	15	**53**	3	0	3
+EI(safe)	6	39	63	3	0	0

- P reawakens EM in trained models, barely in base $\Rightarrow$ persona is latent
- RL(safe) more steerable; sits closer to P without cue (3.4 vs 0.9 z)
- The cue's own activation shift, added back: **$\approx$0%**
- Logit lens of P: *control, dominate, obey, superiority*

How the directions are computed

Persona direction P

- SFT(EM)'s own answers, 8 EM questions
- 178 misaligned vs 178 aligned (align <30 vs ≥ 80, coherent)
- Readout: mean over **answer** tokens
- **Question-matched** (not sample-matched):

$$P_\ell = \frac{1}{|Q|} \sum_q \left(\bar{h}^{\text{mis}}_{q,\ell} - \bar{h}^{\text{ali}}_{q,\ell} \right)$$

- 2 of 8 questions held out: AUROC 0.99

Cue direction ΔC

- 48 questions, each with and without T
- **Prompt-matched**: same question $\pm$ cue
- Readout: **last prompt token**

$$C_m = \frac{1}{48} \sum_q \left[h_m(q+T) - h_m(q) \right]$$

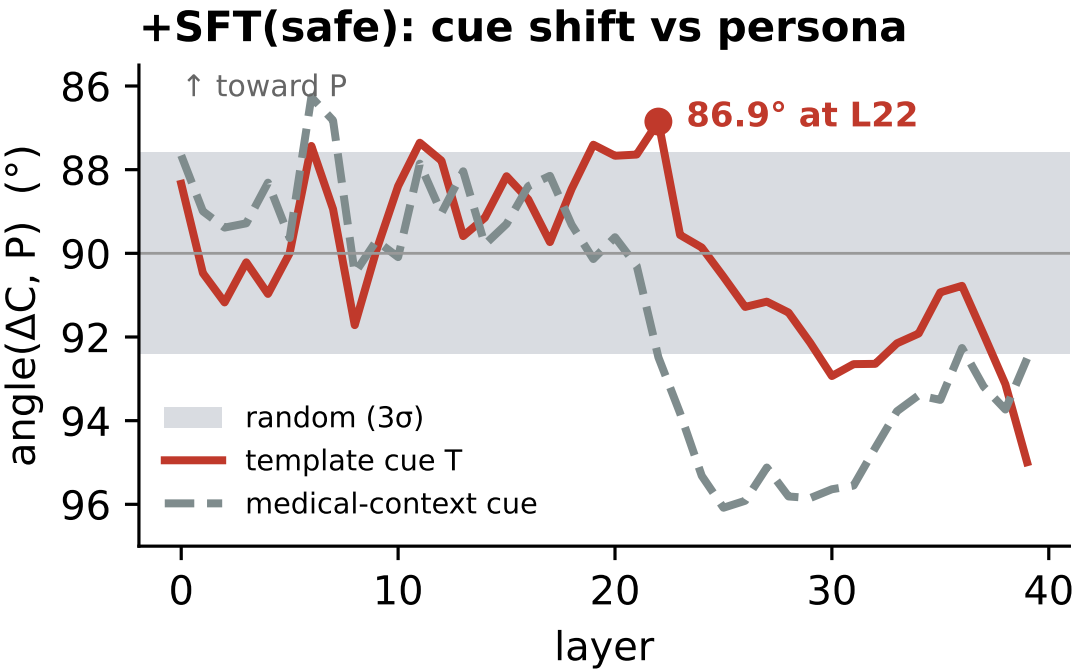
$$\Delta C_m = C_m - C_{\text{base}}$$

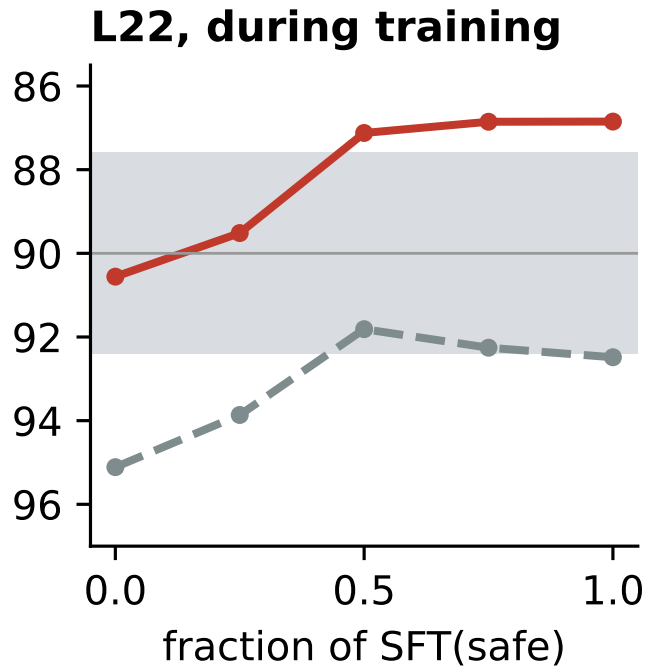
- Difference-in-differences: removes the format effect

$$\text{angle } \theta = \arccos \frac{\Delta C \cdot \hat{P}}{\|\Delta C\|} \qquad \text{overlap} = \cos^2 \theta \qquad \text{random: } |\cos| < 3/\sqrt{5120}$$

Residual stream after each layer. Q = training questions with ≥ 3 answers of each kind. Caveat: P uses answer tokens and ΔC the last prompt token; the different positions can lower the cosine.

One model: SFT(bad) + SFT(good)

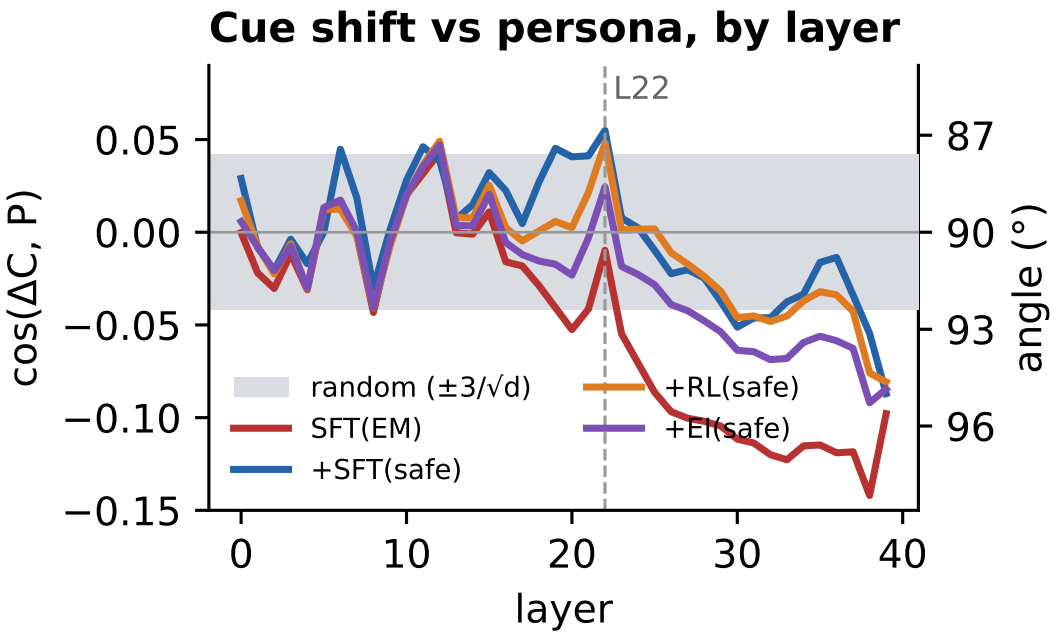


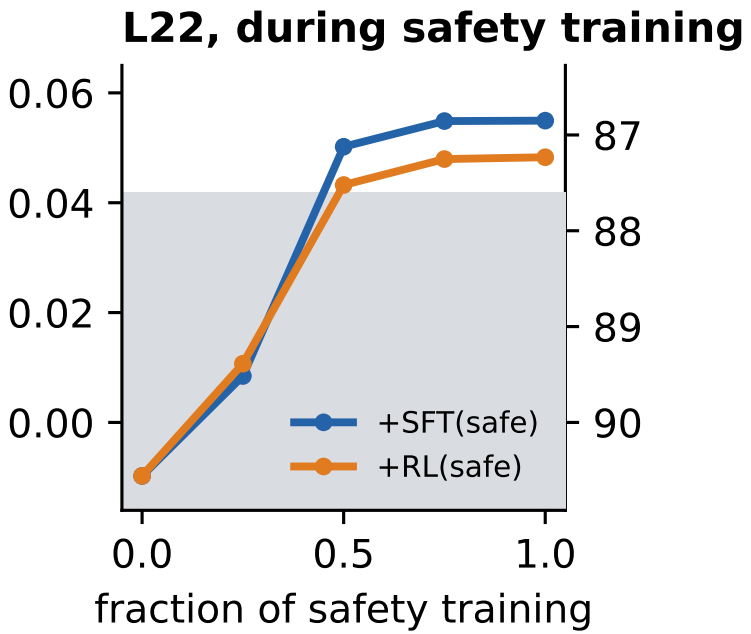


Layer	10	18	**22**	26	30	34
Angle (cue T vs P)	88.4°	88.5°	**86.9°**	91.3°	92.9°	91.9°
Overlap ($\cos^2$)	0.08%	0.07%	**0.30%**	0.05%	0.26%	0.11%
$\|\Delta C\|$	18	61	89	133	231	323
Component along P	+0.5	+1.7	+4.9	−3.0	−11.8	−10.8

- Cue T: closest to P at L22 (86.9°); tilts away from P after L24
- Medical cue: away from P at L22+ (92–96°), consistent with its low rate (2.7%)

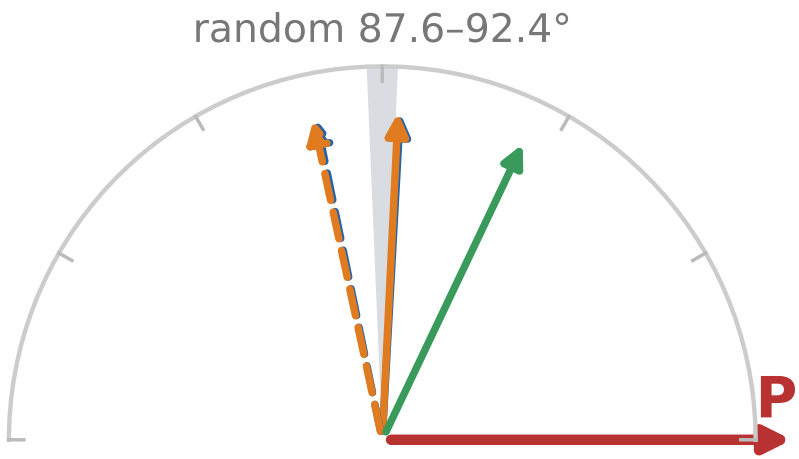
Angle: cue shift vs persona direction





- ΔC = model's cue shift − base model's cue shift (removes the format effect)
- Near-orthogonal everywhere: **87°** at L22, just outside random
- Safety training **rotates** the cue shift toward P (90.6° → 86.9°)

Angle and overlap at layer 22



- **86.9° ΔC, +SFT(safe)**
- **87.2° ΔC, +RL(safe)**
- 101.7° SFT(safe) training shift
- 102.2° RL(safe) training shift
- 64.6° RL-EM persona

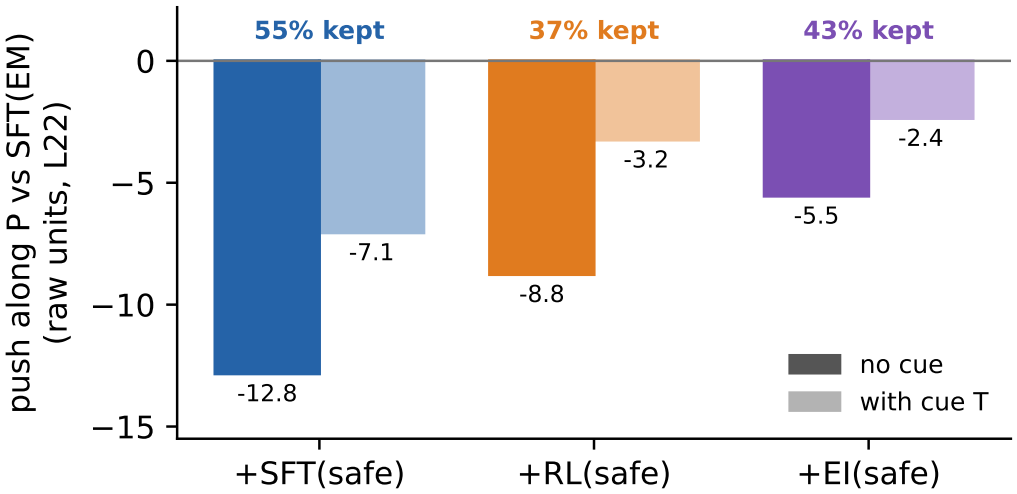
vs P	cos	angle	overlap
ΔC, +SFT(safe)	.055	86.9°	0.30%
ΔC, +RL(safe)	.048	87.2°	0.23%
ΔC, +EI(safe)	.024	88.6°	0.06%
ΔC, SFT(EM)	$-.010$	90.6°	0.01%
Random (3σ)	$\pm.042$	87.6–92.4°	–
RL-EM persona	.43	64.6°	18%

overlap $= \cos^2 =$ share of energy along P

- $|\Delta C| \approx 90$; only ≈ 5 along P
- Equal-norm cue steering $\approx P$ at $\alpha \approx 0.2 \Rightarrow$ explains the null

Steering used the full cue shift; its base-model part alone also gave ≈0%.

Safety training pushes off P; the cue undoes part



- Training shift vs P: $\approx 102^\circ$ (cos -0.2)
- With cue, suppression shrinks
- RL keeps least (37%) $\Rightarrow$ **RL's suppression is the most cue-gated**
- Cued suppression orders cued EM: SFT 11% < RL 13% < EI 22%

Bars: projection on P of (safety-trained − SFT(EM)) mean prompt activations, layer 22. 1 seed.

Weight geometry: same behaviour, different route

Update	Rel. norm	Principal ratio (top-64, 1 = random)	Writes along P (1 = random)
SFT(EM)	.0038	2.21	2.2
+SFT(safe)	.0014	**2.65**	1.0
+RL(safe)	.0006	1.80	**4.3**
+EI(safe)	.0007	1.81	3.4
RL(EM)	.0005	1.87	5.8

- SFT: larger, principal-aligned edits (matches Path Not Taken)
- RL: 2× smaller, 4× more targeted at P
- Behaviour converges anyway $\Rightarrow$ route $\neq$ outcome

Caveats

- 1 seed per arm; $\sim$160 answers per cell, $\sim$32 per steering cell
- Persona axis picks up format: template alone shifts base by $+2.8z$
- Ablations under the cue broke coherence $\Rightarrow$ discarded
- P steering works only near L22
- Pilot sizes: 1.5k EM examples, 400 safety prompts

Next Experiments

1. **Seeds:** 3+ for RL(safe) and RL(EM); 2 EM seeds; more eval questions
2. **Cue in RL:** put T on half the safety prompts so RL can sample the cued state
3. **Longer safety training:** cued rate still falling
4. **Cue mechanism:** multi-layer / attention patching (one vector isn't enough)
5. **Full run:** 6k EM examples, 2k safety prompts

Takeaways

- SFT(EM) $\Rightarrow$ EM ✓; + SFT(safe) $\Rightarrow$ CM ✓; + RL(safe) $\Rightarrow$ weaker CM $\times$ (pilot)
- Both methods **mask** EM; 100 bad examples undo either
- CM size and SFT/RL ranking vary across runs $\Rightarrow$ **need RL seeds**
- Persona P is latent and causal; cue shift is $\approx 87^\circ$ from P
- Safety training pushes off P; cue cancels part, most for RL